

Automation with Heterogeneous Agents: the Effect on Consumption Inequality

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February 23, 2024

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Abstract

I study technological change as a candidate for the increase in consumption inequality in the United States. I build an incomplete market model with educational choice combined with a task-based model on the production side. I consider two channels through which technology affects inequality: the skill an agent can supply in the labor market and the level of capital she owns. I show that the model replicates the increase in consumption inequality between 1981 and 2008 and that educational choice and the return to wealth are quantitatively important in explaining the rise in consumption inequality.

Keywords: Automation, Technological Change, Tasks, Productivity, Inequality, Factor Shares.

JEL Classification: D63, O33, O14, O31, J23, J24.

I thank Javier Fernandez-Blanco, Jordi Caballé, Árpád Ábráhám, Ufuk Akcigit, Christopher Busch, Jan Eeckhout, Philipp Kircher, Alexander Ludwig, Albert Marcet, Javier Miranda, Steffen Müller, Francesc Obiols-Homs, Luis Rojas, Alessandro Ruggeri, Raul Santaaulalia-Llopis and Eugenia Vella for their support, comments, and suggestions. E-mail: tommaso.santini@iwh-halle.de. Website: www.santinit.com

1 Introduction

Since the beginning of the eighties, consumption inequality in the United States has risen.¹ Depending on how consumption is estimated, the increase in inequality ranges from 35% to a staggering 95%, signaling a profound societal shift.² In this paper, I evaluate the role that technological change, and in particular the automation of tasks, might have played in the observed increase in consumption inequality.

To this end, I build a general equilibrium model combining two theoretical frameworks. On the household side, I employ an Aiyagari incomplete market model with educational choice while, on the production side, I use a task-based model borrowed from [Acemoglu and Restrepo \(2018b\)](#). At the beginning of their lives, agents decide whether to pursue a college education or not. Following that, agents face skill-specific uninsurable idiosyncratic risk and choose how much to save and consume. A unique final good is produced by aggregating a unit measure of tasks. Three inputs of production, capital, unskilled labor, and skilled labor are endogenously allocated to perform tasks, given their productivity and endogenous factor prices.

Automation directly displaces low-skill workers from some of the tasks they perform—displacement effect—and increases aggregate productivity—productivity effect. The net effect on the wage of the low-skill workers depends on the trade-off between the displacement and the productivity effect. Moreover, as high-skill workers perform tasks that cannot be performed by machines, automation increases the relative demand for skilled relative to unskilled workers ([Acemoglu and Autor, 2011](#)). I consider the effect of the introduction of new tasks in which labor has a comparative advantage, which has been argued to be one of the most important forces countervailing the displacement effect of automation.³ In my model, the introduction of new

¹See [Aguilar and Bilal \(2015\)](#), [Attanasio, Battistin, and Ichimura \(2004\)](#), [Attanasio and Pistaferri \(2014\)](#), [Attanasio and Pistaferri \(2016\)](#), and [Heathcote, Perri, and Violante \(2010\)](#).

²[Attanasio and Pistaferri \(2016\)](#) show that consumption inequality, measured as the variance of log-consumption, varies between 30% to 95%. The variation in this result depends on the different ways in which consumption is computed in the data.

³Automation technologies might need labor in order to be operated and this, directly, can create demand for new jobs (or new tasks for existing jobs). Moreover, by decreasing the relative price of labor with respect to capital, automation incentivizes the development of new, labor-intensive technologies. [Acemoglu and Restrepo \(2019\)](#) show that the introduction of new

tasks increases productivity and the demand for high-skill relative to low-skill workers. The assumption that new tasks increase the relative labor demand of more educated workers receives support from the data.⁴

Technological change affects labor demand and wages, but also the return to capital. In particular, the adoption of automation technology increases capital demand and, therefore, the return to wealth for the agents who lend capital to the firms.⁵ As wealth is unequally distributed among the agents, the increase in the price of capital has distributional implications. In this model, automation increases *total income* inequality for two different but interconnected reasons. First, because it increases labor income inequality due to differential ways in which technology impacts skill demand. Second, because it increases capital income inequality by raising the return to wealth. As agents who earn a higher salary are also able to accumulate more wealth, the rise in the return to wealth *increases* total income inequality.⁶

The widened gaps in the income distribution translate into increased consumption inequality. The mapping between these two distributions depends on the aggregation of individual saving decisions of the agents which, in turn, depends on the amount of risk agents face. I assume that labor income risk fluctuations depend on education, as this is the case in the data (Güvenen, 2009).

This paper concentrates on the analysis of consumption and consumption inequality with the overarching objective of establishing a comprehensive framework to investigate the dynamic repercussions of automation on welfare. In comparison to income, consumption serves as a more precise indicator of the impact of automation on the population's living standards. Consumption behaviors, in particular, offer a nuanced perspective because they incorporate future

tasks can account for around 50% of the employment growth in the US between 1980 and 2010. Over the same period, a 10 log points increase in labor demand is attributed to the introduction of new tasks.

⁴Acemoglu and Restrepo (2018b) show that occupations with a greater number of new job titles employ, on average, workers with more years of schooling.

⁵While the return on risk-free assets (treasury bonds) has been going *down* in the United States, the return on the entire US capital stock - which is the relevant measure determining inequality (Moll et al., 2019) - has been *increasing* since the eighties (Gomme et al. (2011), Caballero et al. (2017)).

⁶Gomes et al. (2018) consider a different interesting channel through which automation can directly affect the wealth distribution: workers in robot-exposed industries rebalance their financial portfolio away from the stock market, which decreases the average return they can attain.

expectations. As I will show, this becomes especially pertinent in the context of an automation displacement shock, where the beneficial effect of the productivity boost associated with automation takes time to materialize.

Finally, I model educational choice to allow agents to react by adjusting their skill supplies as technology impacts relative skill demand. Ignoring this channel would result in an overestimation of the effect of automation on the education premium since the education decision buffers the increase in the premium.

I calibrate the model to the US economy over the period between 1978 and 1981, paying particular attention to the comparative advantage structure of the inputs of production. With the calibrated model, I compute transitional dynamics to a new steady-state with different levels of technology. Specifically, I show that after an automation shock, the wages of workers without education decrease in the short run and recover along the transition. Despite the short-run decrease in low-skill labor income, total income does not decline for every agent without education. Indeed, wealthier low-skill agents can experience a rise in total income, as the increase in the return of capital compensates for the decline in labor income.

Thereafter, I estimate from the data measures of task automation and task introduction spanning from 1981—the initial steady-state—to 2008. I plug these series into the model and compute the implied transitional dynamics. I show that the estimated technological change explains the increase in consumption inequality observed in the US in the years under study. Moreover, the model explains around 35% of the increase in the college premium and around 53% of the increase in the share of workers with a college degree in the US. Finally, I estimate the role of various components in the model in explaining the increase in inequality. In particular, I show that both the return to wealth channel and the endogenous education decision are quantitatively important. To estimate the role played by the increase in the return to wealth, I compute the transition between steady-states with the interest rate fixed to its initial steady-state value. In this case, the effect of technological change on inequality is 4% lower as it increases only the education premium and it does not directly increase inequality in capital income. To quantify the role of endogenous educational choice, I compute the transition with the probability of dying equal to zero. In this way, the shares of educated and uneducated workers remain fixed to the initial steady-

state value. With this decomposition, I show that without allowing the agents to adjust their skill supplies as the demand changes, the overall effect of technology on inequality is twice as large.

The rest of the paper is organized as follows: the remaining part of the introduction discusses this paper's relation with the existing literature. In Section 2, I explain the structure of the model. In Section 3.1, I take the model to the data. In Section 3.2, I discuss the mechanisms of the model. In Section 3.3, I describe the estimation of the technological sequences that I use in Section 3.4 to contrast the model with the data. In Section 3.5, I quantify the role of various elements of the model. Section 4 concludes.

Literature Review

First, this paper contributes to the literature that focuses on the determinants of income and consumption inequality. There are several papers that use the [Aiya-gari \(1994\)](#), [Bewley \(1986\)](#), and [Huggett \(1993\)](#) framework to study the role of technology on inequality as, for instance, [Heckman, Lochner, and Taber \(1998\)](#), [Hubmer, Krusell, and Smith Jr \(2016\)](#), and [Kaymak and Poschke \(2016\)](#). The most important differences with respect to these studies are the way in which I conceptualize technological change—a combination of automation and creation of new tasks—and that I consider the effect of the return to wealth on inequality. Methodologically, my paper is similar to [Kaymak and Poschke \(2016\)](#) in the way I decompose the effect on consumption inequality of various channels in the model. Another paper that underlines the importance of the return to wealth is [Moll, Rachel, and Restrepo \(2019\)](#). They combine a task-based model with a perpetual youth structure with imperfect dynasties, which is another way to obtain a determinate wealth distribution and study the implication of the return to wealth on inequality. Using instead an incomplete market model, I can account for differences in labor income risk between education groups ([Guvenen, 2009](#)). Another key difference with respect to that paper is that I do model educational choice, and I show that it is crucial in the observed increase in inequality. Finally, I also consider the effect of the introduction of new tasks.

Second, this paper contributes to the literature that studies the effect of automation, spurred by recent advances in technological capabilities. Like the majority of papers in this literature, I use a task-based model of production ([Zeira](#)

(1998), [Acemoglu and Autor \(2011\)](#) and [Acemoglu and Restrepo \(2018b\)](#)). Most of the papers studying the effect of automation using general equilibrium models either focus on aggregate labor demand (as [Acemoglu and Restrepo, 2018b](#)) or consider labor income inequality ([Hémous and Olsen, 2014](#)). But as automation increases the demand for capital, I show that the return to wealth channel is quantitatively important. I extend the conceptual models used in previous research by including heterogeneous capital accumulation that depends on labor income and individual risk. In a representative household model (e.g. [Acemoglu and Restrepo, 2018b](#)) automation always increases total income and consumption. That is because automation always increases output; hence, even if wages are reduced, the household is compensated with a higher capital income. This is not true for all the agents in a model with endogenous wealth distribution. Finally, [Beraja and Zorzi \(2022\)](#) combine a dynamic discrete occupational choice model with a heterogeneous agents framework. They show that the equilibrium is constrained inefficient as firms fail to internalize the consequences of worker displacement.

2 Model

The model combines two theoretical frameworks. The consumption side borrows the basic features of the [Aiyagari \(1994\)](#) incomplete market model combined with educational choice. The production process, instead, is modeled with a task-based production function borrowed from [Acemoglu and Restrepo \(2018b\)](#).

The population is normalized to one and time is discrete and indexed by t . I make the dependence on time explicit to underline non-stationarity in the model. An agent is born with a level of assets and chooses whether or not to become high-skill by paying a cost $\theta(a)$ which is a decreasing function of asset holdings. This decision is permanent for an agent. Then, during her life, the agent chooses how much to save and how much to consume. She cannot borrow, and her labor supply is exogenous. In every period there is a probability of dying d ; when an agent dies, her offspring inherits the level of asset holdings the agent had in her last period of life.

Productivity differs between skill groups but is identical within each skill

group. Consequently, wages differ between skills but are identical within skills. On top of this, agents face a non-insurable idiosyncratic shock and this creates heterogeneity within each skill group.

Consumption Side:

The initial problem of an agent is the following:

$$v_t^n(a) = \max \left\{ \mathbb{E}_{\varepsilon^h} \left\{ v_t^h(a, \varepsilon^h) \right\} - \theta(a), \mathbb{E}_{\varepsilon^\ell} \left\{ v_t^\ell(a, \varepsilon^\ell) \right\} \right\}, \quad (1)$$

where $v_t^n(a)$ is the value function of new-born agents, $v_t^h(a, \varepsilon^h)$ is the value function of a high-skill agent that depends on the asset holdings and on the realization of the labor endowment shock ε^h .⁷ Similarly, $v_t^\ell(a, \varepsilon^\ell)$, is the value function of a low-skill agent. The value of becoming a high-skill worker is reduced by a fixed cost, $\theta(a)$, which is a decreasing function. The expectations are formed using the stationary Markov distributions which are type-specific.

Once the decision regarding the type is taken, agent i solves the following problem, with $j = \{\ell, h\}$:

$$v_t^j(a_{i,t}, \varepsilon_{i,t}^j) = \max_{c_t, a_{t+1}} \left\{ u(c_{i,t}) + \beta(1-d) \sum_{\varepsilon_{t+1}^j} \pi(\varepsilon_{t+1}^j | \varepsilon_{i,t}^j) v_{t+1}^j(a_{i,t+1}, \varepsilon_{i,t+1}^j) \right\}, \quad (2)$$

$$\text{subject to } c_{i,t} + a_{i,t+1} = (1 + r_t - \delta)a_{i,t} + w_t^j \cdot \varepsilon_{i,t}^j, \quad \text{and } a_t > 0,$$

where w_t^j is the type-specific wage rate, and $\varepsilon_{i,t}^j$ is the idiosyncratic shock, which follows a type-specific Markov process. r_t is the interest rate and δ is capital depreciation.

Production Side:

The production side of the economy follows the task-based model with capital and labor with two types of skills in [Acemoglu and Restrepo \(2018b\)](#). My production framework, however, has two important differences. First, I relax

⁷I refer to the shock as “labor endowment” as in [Aiyagari \(1994\)](#). In the literature is perhaps more common to use “productivity shocks”. However, in this model, the shock does not change the productivity of the agents which is a very precise object (see below). The productivity of the agent determines her wage rate, while, the shock, together with the wage rate determines the total labor income.

their assumption of fixed shares of worker types by introducing the education decision. Second, I change the comparative advantage structure of the inputs of production in order to facilitate the calibration in Section 3.1.

There is a unique final good produced with a continuum of tasks,

$$\ln Y = \int_{N-1}^N \ln[y(x)] dx, \quad (3)$$

where Y is the output of the final good and $y(x)$ is the quantity of task x produced. The final good is produced with a unit measure of tasks that ranges from $N - 1$ to N . An increase in N , which corresponds to the introduction of new tasks, does not alter the total measure of tasks in the economy. Each task is produced with a linear production function,

$$y(x) = \gamma_m(x)m(x) + \gamma_\ell(x)l(x) + \gamma_h(x)h(x), \quad (4)$$

where $m(x)$ is the amount of capital (machines) used in the production of task x . The function $\gamma_m(x)$ is the productivity schedule of capital, which is a function that for every task gives the productivity of capital in that task. The same holds for low-skill and high-skill labor.

The relationships between the productivity schedule of capital and the two productivity schedules of labor determine, together with factor prices, how the production of tasks is split between capital and labor. I assume the following about these relationships:

Assumption 1

$$\frac{d}{dx} \left(\frac{\gamma_\ell(x)}{\gamma_m(x)} \right) > 0 \quad \text{and} \quad \frac{d}{dx} \left(\frac{\gamma_h(x)}{\gamma_m(x)} \right) > 0. \quad (A1)$$

The ratio of the productivity schedule of the two types of labor over the productivity schedule of capital is an increasing function in x . Therefore, labor has a comparative advantage in higher-indexed tasks with respect to capital. Regarding the relationship between the productivity of the two types of labor I assume,

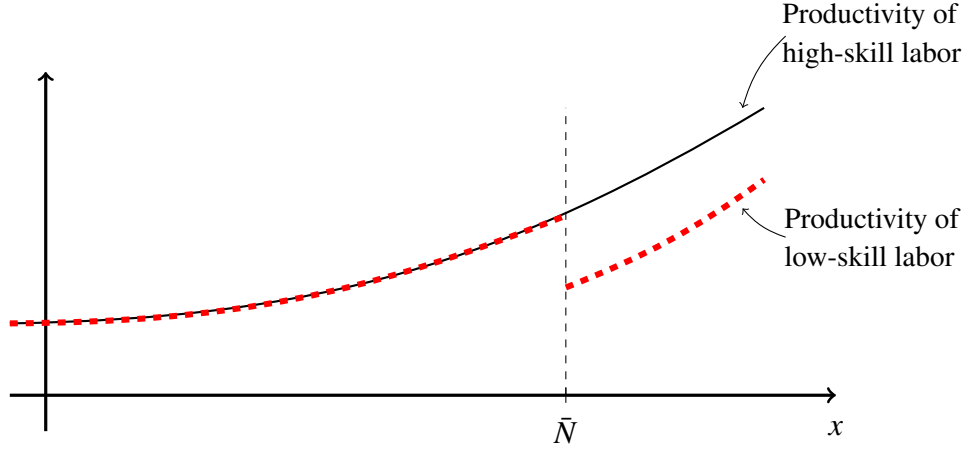


FIGURE 1: The solid black line represents the productivity schedule of high-skill labor in the space of tasks. The red dotted line represents the productivity schedule of low-skill labor. The two functions are *equal* for $x < \bar{N}$.

Assumption 2

$$\gamma(x) = \begin{cases} \gamma_h(x) & x \leq \bar{N} \\ \gamma_h(x) \cdot \Gamma & x > \bar{N}, \end{cases} \quad (\text{A2})$$

where Γ is a parameter smaller than 1. Figure 1 provides a visual representation of the two productivity schedules of labor. The two functions are *equal* for $x < \bar{N}$, while the low-skill productivity is scaled down by a factor equal to Γ for tasks with an index higher than $\bar{N}$. This implies that high-skill labor has a comparative advantage in higher-indexed tasks with respect to low-skills. The threshold $\bar{N}$ can be thought of as a division between old and new tasks, or complex and non-complex tasks. I model the comparative advantage structure between the two types of labor with a discontinuity for parameter identification purposes. As I explain in detail in the calibration section (Section 3.1), this functional form allows me to calibrate the exact shape of the productivity schedules. In particular, in this way, I can discipline the trade-off between the displacement and productivity effects associated with the adoption of automation technology.

The highest indexed task produced with capital, $\tilde{I}$, is given by solving the following equation:

$$\tilde{I}: \frac{r}{\gamma_m(\tilde{I})} = \frac{w_\ell}{\gamma_\ell(\tilde{I})}, \quad (5)$$

hence, $\tilde{I}$ is the task in which the effective costs—price over productivity—of

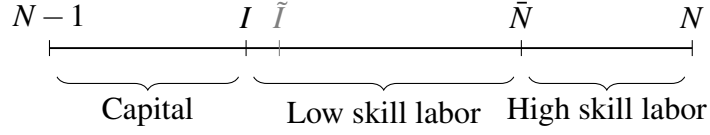


FIGURE 2: Allocation of factors of production in the unit measure of tasks.

capital and labor are equal.

The division between high and low skills in the labor set follows a similar logic. However, given the discontinuity in the productivity of the low-skills, there is not one clear equation that pins down the separation threshold, as we have for the division capital-labor. Therefore, for simplicity, I make the following assumption:⁸

Assumption 3

$$\Gamma < \frac{w_\ell}{w_h} < 1, \quad (\text{A3})$$

which, in turn, implies the following:

$$\begin{aligned} \frac{w_\ell}{\gamma_\ell(x)} &< \frac{w_h}{\gamma_h(x)} & \text{if } x < \bar{N}, \\ \frac{w_\ell}{\gamma_\ell(x)} &> \frac{w_h}{\gamma_h(x)} & \text{if } x > \bar{N}. \end{aligned}$$

The left-hand side of both equations is the effective cost of producing tasks with low-skill labor while the right-hand side is the same variable for high-skills. These equations tell us that when assumptions A2 and A3 are satisfied, only high-skill workers are employed in *new* tasks, and only low-skills are employed in *old* tasks. The separation threshold between the two types of labor is equal to $\bar{N}$. Assumptions A1, A2, and A3 imply that the unit measure of tasks is divided into three areas: tasks performed by capital, tasks performed by low-skill labor, and tasks performed by high-skill labor (see Figure 2).

I now explain how automation is modeled in this framework. As said, $\tilde{I}$ is the highest indexed task that is optimal to automate given productivity schedules and factor prices. I now define I as the highest indexed task that is *feasible* to automate. This means that for $x > I$, the technology that allows producers to use machines to perform these tasks simply does not exist. In general, then, the

⁸This assumption is made on endogenous objects and must be verified ex-post in equilibrium.

highest indexed task automated in equilibrium, I^* , is equal to

$$I^* = \min\{I, \tilde{I}\}.$$

For some combinations of the parameters in the model, the profit maximization problem of the firm is constrained in the equilibrium, that is $I^* = I$. In these cases, automation is a *relaxation* of this constraint, an increase in I . For the rest of the paper, I focus on this case because it is the only one that allows me to study the implication of an invention in automation technology. Indeed, when some automation technologies are not adopted ($\tilde{I} < I$), an increase in I has absolutely no effect on the equilibrium in this model.

The total output, Y , can be rewritten as a Cobb Douglas in the three factors of production, in which, crucially, factors' shares are endogenous and depend on technology.⁹

$$Y = G \left(\frac{K}{I - N + 1} \right)^{I - N + 1} \left(\frac{L}{\bar{N} - I} \right)^{\bar{N} - I} \left(\frac{H}{N - \bar{N}} \right)^{N - \bar{N}}, \quad (6)$$

where total factor productivity, G , is equal to,

$$G = \exp \left(\int_{N-1}^I \ln(\gamma_m(x)) dx + \int_I^{\bar{N}} \ln(\gamma_\ell(x)) dx + \int_{\bar{N}}^N \ln(\gamma_h(x)) dx \right).$$

Notice how total factor productivity depends on the productivity schedules and on the level of technological progress, I and N . The expressions of factor prices take the usual form,

$$r = Y \cdot \frac{I - N + 1}{K}, \quad (7)$$

$$w_\ell = Y \cdot \frac{\bar{N} - I}{L}, \quad (8)$$

$$w_h = Y \cdot \frac{N - \bar{N}}{H}. \quad (9)$$

The price of each factor is proportional to total output and to the share of the factor in aggregate production, and inversely proportional to the supply of the

⁹For a detailed derivation of the production function in (6) with one type of labor, the interested reader is referred to [Acemoglu and Restrepo \(2018a\)](#).

factor.

Before turning to the definition of equilibrium, it is useful to describe the transition of the distribution between a generic period t and $t + 1$. In every period t , each agent is characterized by three variables, the level of assets she owns, $a_{i,t}$, the Markov state, $\varepsilon_{i,t}^j$, and her education level, e_i . At the end of the period, a random sample of size d —which corresponds to the probability of dying in t —is drawn from the agents' distribution over states at time t , λ_t . Before the beginning of the next period, the deceased agents are replaced by their offspring who inherit their level of capital. First, they decide their level of education based on (1). Second, their Markov state is realized based on the education-specific stationary Markov distribution. The transition of the agents who do not die in period t to a new position in state space in $t + 1$ depends on the solution of the value function (2)—that depends on their level of asset holdings, Markov state, and education type—and on the realization of the idiosyncratic shock in $t + 1$.

Equilibrium:

Given the two sequence describing technological change, $\{I_t\}_{t=0}^\infty$ and $\{N_t\}_{t=0}^\infty$, a recursive competitive equilibrium are sequences of value functions $\{v_t^h\}_{t=0}^\infty$ and $\{v_t^\ell\}_{t=0}^\infty$, policy functions $\{c_t^h, a_{t+1}^h\}_{t=0}^\infty$ and $\{c_t^\ell, a_{t+1}^\ell\}_{t=0}^\infty$, firm's choices $\{L_t, H_t, K_t\}_{t=0}^\infty$, prices $\{w_t^\ell, w_t^h, r_t\}_{t=0}^\infty$, and distributions $\{\lambda_t\}_{t=0}^\infty$ such that, for all t ,

- Given prices, the policy functions solve the agents' problems and the associated value functions are $\{v_t^h\}_{t=0}^\infty$ and $\{v_t^\ell\}_{t=0}^\infty$.
- Given prices and technology, firms choose optimally labor inputs and capital.
- The labor markets clear,

$$H_t = \left[\left(\Pi^{*,h} \right)^T \cdot \varepsilon^h \right] S_t^h,$$

$$L_t = \left[\left(\Pi^{*,\ell} \right)^T \cdot \varepsilon^\ell \right] S_t^\ell,$$

where $\Pi^{*,j}$ is the stationary distribution associated with the Markov pro-

cess of type j , ε^j is a vector containing the values of the shock corresponding with the stationary distribution and S_t^j is the mass of agents that belong to type j at time t .

- The asset market clears,

$$K_t = \int_{A \times E} a_{t+1}(a_t, \varepsilon_t) d\lambda_t,$$

where A and E are the spaces of asset holdings and labor endowment shocks ε , respectively.

3 Quantitative Results

I now turn to the quantitative analysis of the model. First, I explain how I calibrate the model parameters. Then, using the calibrated model, I discuss the mechanisms at play. To this end, I separately report the transitional dynamics between steady-states for two different shocks: (i) an increase in automation and (ii) an introduction of new tasks in which labor has a comparative advantage. For the sake of clarity, I assume that these shocks are instantaneous, meaning the final steady-state value of the shock is reached immediately. This approach makes it easier to understand the reaction of the model economy compared to a case in which the shock happens gradually. This strong assumption is relaxed in the main result of the paper, in which I compute the transitional dynamics implied by sequences of technological parameters estimated from the data. Finally, through a decomposition exercise, I quantitatively evaluate the contribution of various components of the model in determining the increase in inequality.

3.1 Bringing the model to the data

To inform my decision regarding in which period to assume that the US economy was in a steady state, I refer to the composition-adjusted college premium, which I report in Figure 3. The data is taken from the March CPS database; I restrict the sample to include only full-time full-year workers aged between 16 and 64. This measure of the college premium, taken from [Acemoglu and](#)

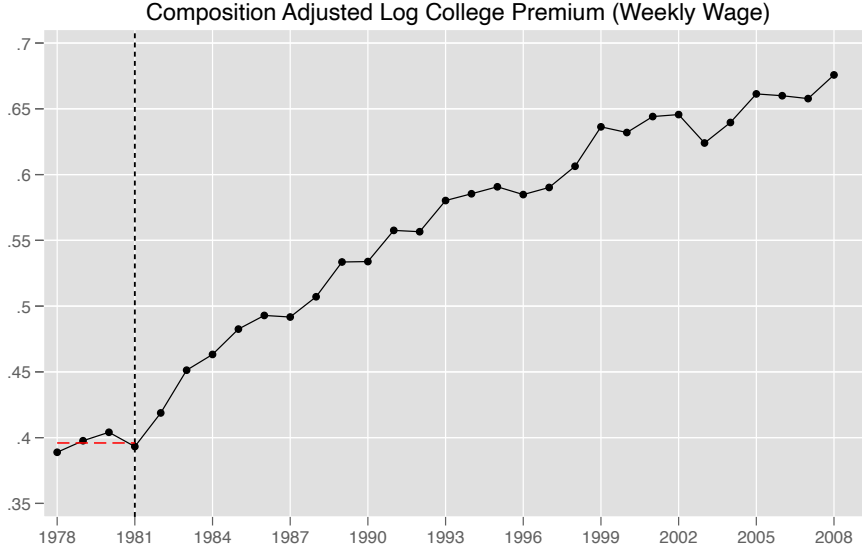


FIGURE 3: Composition-adjusted log college premium from 1978 to 2008 in the United States. Data from March CPS, full-time, full-year workers. The steady-state value—the red dashed line in the graph—is equal to 0.39. I use the estimation technique of [Acemoglu and Autor \(2011\)](#).

[Autor \(2011\)](#), ensures that this statistic is not “mechanically affected by shift in experience, gender composition, or average level of completed schooling within the broader categories of college and high-school graduates”. I assume that the economy is in a steady state from the year 1978 to 1981—red dashed horizontal line in the graph—because in this period the college premium displays more stability relative to the whole period.

In Table 1, I report the calibrated parameters and the relative targets or sources used in the calibration. The table is divided into two parts: preferences and technology. The result of the calibration of the labor income risk parameters is excluded from the table but is also discussed in this section. I now explain the reasoning behind the calibration strategy for each parameter. Regarding the parameters that enter the preferences of the agents, I set the dying probability equal to 3% to imply an average working life of 33 years. Regarding the education cost in (1), I assume the following functional form:

$$\theta(a) = \frac{\tilde{\theta}}{a}.$$

DESCRIPTION	VALUE	TARGET/SOURCE
<i>PREFERENCES</i>		
σ Risk Aversion	2	Standard
β Discount	0.95	Standard
δ Depreciation	6%	Standard
d Death probability	3%	33 years average working life
$\tilde{\theta}$ Education Cost	15.04	Share of workers with col. degree
<i>TECHNOLOGY</i>		
N Highest-indexed task	1	Normalization
I Highest-indexed automated task	0.35	Labor share = 0.66
$\tilde{\gamma}$ Productivity	0.12	$K/Y = 3$
q_y Productivity of labor 1	0.7	Cost saving = 30%
m Productivity of labor 2	1.66	$\hat{z} = 1.67$
$\tilde{N}$ Highest-indexed task non-college	0.84	Log college premium = 0.43

TABLE 1: Calibrated parameters of the model.

To estimate the parameter $\tilde{\theta}$, I use the simulated method of moments targeting the average share of workers with a college degree in the US between 1963 and 1981, which is equal to 14%. The remaining preference parameters have standard values taken from the literature. Labor income risk is calibrated using values from Guvenen (2009). Taking data from the Panel Study of Income Dynamics (PSID) covering 1968 to 1993, he estimates an AR(1) income process separately for college and non-college graduates.¹⁰ The values he estimates for the persistence and the variance of the innovation are reported in Table 2. I compute the associated Markov process using Tauchen’s method. I therefore need to set the number of Markov states, S , and the maximum number of standard deviations from the mean, i.e. the dispersion of the Markov’s state space. I set $S = 9$ and the maximum standard deviations equal to 1. From the discretization, I obtain the conditional probabilities of the Markov matrix, Π^i , and the vector of Markov states, ε^i . Given that Guvenen (2009) uses log labor earnings to estimate the labor income risk parameters, I have to normalize the values of the Markov process to have $\mathbb{E}(\exp(\varepsilon_i^j)) = 1$. To this end, I symmetrically shift the

¹⁰Using Guvenen words, his sample consists of “[...] male head of households between the ages of 20 and 64. I include an individual into the sample if he satisfies the following conditions for twenty (not necessarily consecutive) years: the individual has (1) reported positive labor earnings and hours; (2) worked between 520 and 5110 hours in a given year; (3) had an average hourly earnings between a preset minimum and a maximum wage rate (to filter out extreme observations). I also exclude individuals who belong to the poverty (SEO) subsample in 1968”.

Parameters	Values
ρ_ℓ	.829
σ_ℓ^2	.022
ρ_h	.805
σ_h^2	.025

TABLE 2: Labor income risk estimates from [Güvenen \(2009\)](#). The persistence, ρ , and the variance of the innovation, σ , of the income processes of workers with and without a college degree which I discretize to obtain the Markov processes.

values found with Tauchen’s method.

I now turn to the discussion of the parameters of the production side of the economy. I normalize the highest-indexed task in the economy $N_0 = 1$ (the sub-index “0” indicates the initial steady-state). To compute the highest indexed task automated in equilibrium I use the following relationship that holds in the model:¹¹

$$I_t = N_t - (\text{LABOR SHARE})_t.$$

The series of the Labor Share (LS) is a crucial object for the results of this paper, as this is used to impute the level of automation in the initial steady-state and, as will be explained in Section 3.3, to estimate the sequences of the technology parameters.

The remaining four parameters $(\tilde{\gamma}, q_y, m, \bar{N})$ in Table 1 determine the shape of the productivity schedules of the factors of production. The functional forms

¹¹From the definition of the labor share,

$$LS = \frac{w_h H + w_\ell L}{Y},$$

substitute the expressions for the wages, (8) and (9), to obtain,

$$LS = \frac{Y(N - \bar{N}) + Y(\bar{N} - I)}{Y} = N - I.$$



FIGURE 4: Labor share. Data from the Bureau of Economic Analysis. The average value of the labor share in the initial steady state is 0.64.

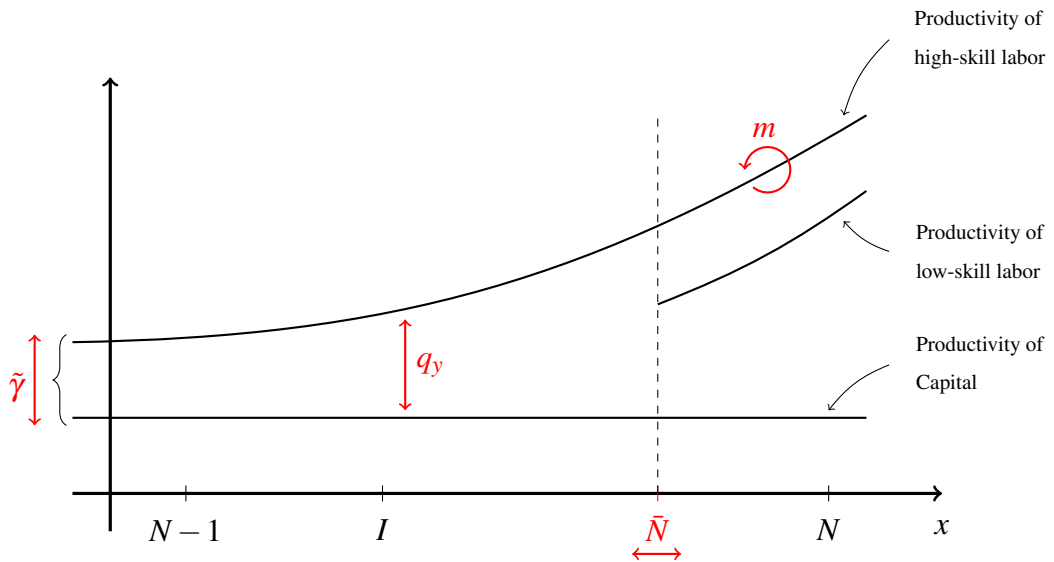


FIGURE 5: Productivity schedules of the inputs of production. In red, are the parameters to calibrate.

chosen for the productivity schedules are the following:

$$\gamma_h(x) = \tilde{\gamma} \cdot q_y \cdot e^{m(x - \frac{I+N}{2})}, \quad (10)$$

$$\gamma_\ell(x) = \begin{cases} \gamma_h(x) & x \leq \bar{N} \\ \gamma_h(x) \cdot \Gamma & x > \bar{N}, \end{cases} \quad (11)$$

$$\gamma_m(x) = \tilde{\gamma}. \quad (12)$$

In Figure 5, I show how the parameters to calibrate affect the shape of the productivity schedules. The parameter $\tilde{\gamma}$ determines the aggregate productivity, as an increase in this parameter shifts up all productivity schedules. The parameter q_y determines the difference between the productivity of capital and that of labor. Finally, $\bar{N}$ controls the difference between the productivity schedule of unskilled and skilled labor while m regulates the slope of the productivity schedule of labor.

As will be more precisely explained in the following section, the effect of task automation on the economy crucially depends on the trade-off between the displacement and the productivity effect. The displacement effect depends on how task automation changes the relative demand for factors, as an increase in I decreases the size of the set of tasks performed by low-skills and increases the size of the set of tasks performed by capital. However, automation also increases productivity and this tends to increase all factor prices. I specify the productivity schedule in (10) so that the parameter m determines precisely this trade-off. As m increases, for a given productivity of capital and a given displacement effect, the cost-saving (and therefore the productivity effect) implied by automation is greater. The chosen specification for the productivity schedule in (10) implies that, in the initial steady-state, a change in the parameter m within a range of values, does not change the equilibrium of the model economy. This is because it does not change the aggregate productivity G and does not change the relative demand for factors.¹² In this way, m is directly linked with the previously

¹²Wages in this model change because of two reasons (i) the aggregate productivity G changes (ii) the relative demand for factors changes. The aggregate productivity does not change with m because, thanks to the specific chosen functional form, as, for instance, m increases, the productivity of skilled workers increases but the productivity of low-skills decreases in a way that perfectly offsets the effect on aggregate productivity. Moreover, the change in productivity does not change the relative demand for labor because, on one side, there is the technological constraint (I) and, on the other side - the threshold that separates the two types of labor - there is the discontinuity of the productivity schedule of the low-skill workers.

mentioned trade-off. The value of m reflects also the difference between the average productivity of workers with a college degree and the average productivity of workers without a college degree. Indeed, as m increases, the average productivity of college workers increases relative to non-college workers. This is perfectly consistent with the relationship between m and the trade-off. As the productivity of the workers who will be automated increases, the cost-saving coming from automation decreases, and the impact of automation on the economy changes. To calibrate this parameter, I use, therefore, the following statistics:

$$\hat{z} = \frac{\text{Average workplace productivity of workers with a college degree}}{\text{Average workplace productivity of workers without a college degree}}.$$

I take the estimate for $\hat{z}$ from [Hellerstein et al. \(1999\)](#), which is equal to 1.67. In that paper, the authors estimate precisely the difference in productivity between workers with and without college education. To build the counterpart of this statistic in the model, I use the implication of assumptions [A2](#) and [A3](#) about the labor productivity schedule. In the initial steady-state, low-skill workers perform tasks between I and $\bar{N}$ and skilled workers between $\bar{N}$ and N . The average productivity of low-skill workers is,

$$\bar{\ell} = \frac{\int_I^{\bar{N}} \gamma_\ell(x) \cdot \frac{L}{\bar{N}-I} dx}{L}.$$

While for high-skill workers,

$$\bar{h} = \frac{\int_{\bar{N}}^N \gamma_h(x) \cdot \frac{H}{N-\bar{N}} dx}{H}.$$

Finally, the moment condition I use to calibrate m is,

$$\frac{\bar{h}(m)}{\bar{\ell}(m)} = \hat{z},$$

which, after plugging the expressions for $\bar{\ell}$ and $\bar{h}$ becomes,

$$\frac{\exp(mN) - \exp(m\bar{N})}{\exp(m\bar{N}) - \exp(mI)} \cdot \frac{\bar{N} - I}{N - \bar{N}} = \hat{z}. \quad (13)$$

This is the reason behind the specific functional form of the schedule for low-skills. Indeed, without the discontinuity, it would be impossible to identify m .

To calibrate q_y I use the estimate from [Acemoglu and Restrepo \(2017\)](#) of the cost-saving associated with the adoption of automation technology. To pin down $\tilde{\gamma}$, I use the capital-output ratio. Finally, for $\bar{N}$, I use the adjusted log college premium, $\widehat{CP}$ (average value over the period from 1963 to 1981, see [Figure 3](#)). Given that $\widehat{CP}$ is the difference between the average log wages for the two education groups, the moment condition is not simply $\log(w_h/w_\ell) = \widehat{CP}$. To explain the derivation of this moment condition, I use the example of a three-state Markov process. The average log wage for low-skill workers is

$$\begin{aligned} \pi_1^{*,\ell} \log(w_\ell \varepsilon_1^\ell) + \pi_2^{*,\ell} \log(w_\ell \varepsilon_2^\ell) + \pi_3^{*,\ell} \log(w_\ell \varepsilon_3^\ell) = \\ \log(w_\ell) \underbrace{(\pi_1^* + \pi_2^* + \pi_3^*)}_1 + (\Pi^{*,\ell})^T \log(\varepsilon^\ell) = \\ \log(w_\ell) + (\Pi^{*,\ell})^T \log(\varepsilon^\ell), \end{aligned} \quad (14)$$

where $\Pi^{*,j}$ is the stationary distribution associated with the type-specific Markov process. Hence, the moment condition is

$$\begin{aligned} \overline{\log(w_h)} - \overline{\log(w_\ell)} = \log(w_h) - \log(w_\ell) + \\ + (\Pi^{*,h})^T \log(\varepsilon^h) - (\Pi^{*,\ell})^T \log(\varepsilon^\ell) = \widehat{CP}, \end{aligned} \quad (15)$$

which, given the model expression of the wage ratio,

$$\frac{w_h}{w_\ell} = \frac{N - \bar{N}}{\bar{N} - I} \cdot \frac{L}{H}, \quad (16)$$

becomes,

$$\log\left(\frac{N - \bar{N}}{\bar{N} - I} \cdot \frac{L}{H}\right) + (\Pi^{*,h})^T \log(\varepsilon^h) - (\Pi^{*,\ell})^T \log(\varepsilon^\ell) = \widehat{CP}. \quad (17)$$

In [Table 3](#), I report the model-generated moments used in the calibration and their data counterparts. All the targeted moments match the data quite well. In [Table 4](#), I report the Gini coefficients of the wealth and consumption distribution of the simulated economy. While these moments were not targeted directly, the model generates 57% of the wealth inequality observed in the data and 62% of the consumption inequality. As nicely summarized by [Castaneda et al. \(2003\)](#), the comparison between the wealth Gini coefficient replicated by my model

Expressions		Data/Targets	Model
$\log(w_h/w_\ell)$	Log coll. premium	0.39	0.39
$\hat{z}$	prod. ratio	1.67	1.67
K/Y	Capital/output	3	2.58
$\frac{w_\ell}{\gamma_\ell(I)} / \frac{r}{\gamma_m}$	Cost saving autom.	30%	30%
S_h	college share	18%	18%

TABLE 3: Calibration results, targeted moment.

Gini coefficients	Data	Model
Consumption	0.24	0.15
Wealth	0.77	0.44

TABLE 4: Untargeted moments. Source of the data counterparts: [Kuhn et al. \(2020\)](#) and [Krueger and Perri \(2006\)](#).

and the actual data aligns with the results of other studies utilizing the Aiyagari framework. Specifically, [Aiyagari \(1994\)](#) finds that, by calibrating the earning process for ex-ante identical households, a Gini coefficient of 0.38 is obtained while the data counterpart is 0.78. Notably, introducing the education dimension with two distinct earning processes enhances the model’s ability to match inequality by 18%. However, my model’s capacity to explain wealth inequality cannot be further improved without the addition of other modeling elements.¹³ It is important to emphasize that this paper does not delve into identifying the primary drivers of wealth dispersion in general; rather, its focus is to shed light on the *overall dynamic effects* of task automation on inequality.

3.2 Discussion of the Mechanisms

To analyze the effect of task automation and the introduction of new tasks, I now report, separately, one transition for each type of technology shock. In Figure 6, I show the transitional dynamics after a permanent and instantaneous increase in the tasks performed with machines (a 5% increase of I).¹⁴ To understand

¹³For instance, [Quadrini \(2000\)](#) adds entrepreneurship, [Krusell and Smith \(1998\)](#) consider discount rate shocks, [Fagereng et al. \(2019\)](#) study the addition of stochastic asset prices and portfolio frictions.

¹⁴This exercise, like all the other exercises in this paper, studies the reaction of the model economy after an “MIT shock”. The agents in the initial steady-state make decisions as

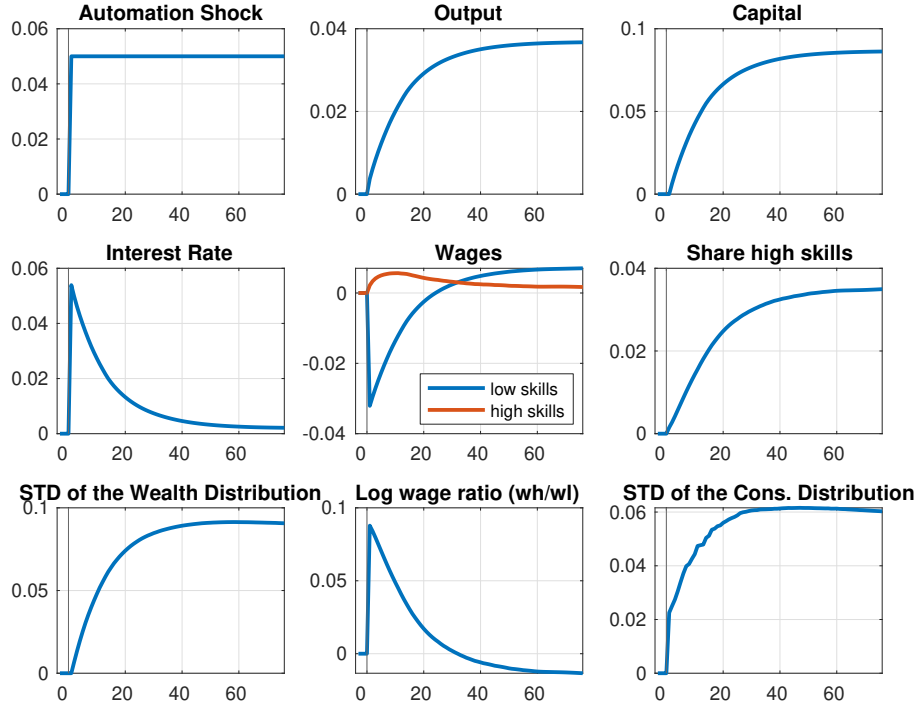


FIGURE 6: Transitional dynamics between the initial, calibrated steady-state and a final steady-state in which the size of the set of tasks performed by capital has increased by 5%. All reported variables are normalized to zero in the initial steady-state.

the reaction of the interest rate, capital, and output, it is convenient to recall equation (7),

$$r = Y \cdot \frac{\uparrow I - N + 1}{K}.$$

In the first period after the shock, the increase in I implies an instantaneous increase in the interest rate. Indeed, the reaction of the aggregate capital stock is sluggish and the supply of capital takes time and does not compensate immediately for the increase in demand. As time goes by, agents start to accumulate more capital and the interest rate decreases until the final steady-state value. As more capital is accumulated, the output increases.

In the short run, the effect of automation on the wage of educated workers is unambiguously positive, as automation increases productivity and does not decrease the demand for educated labor. Instead, for low-skill workers, the effect

if the economy is staying in the steady-state forever. When the shock occurs, the path of aggregate variables is known to the agents immediately.

on the wage in the short run depends on the trade-off between the increase in productivity and the decrease in the number of tasks in which they are demanded by the firms, or, in other words, the fact that they are reallocated to a different set of tasks. This trade-off can be analyzed by taking the following derivative:

$$\frac{d \ln w_\ell}{dI} = \underbrace{\frac{d \ln(Y/L)}{dI}}_{\text{Productivity Effect}} + \underbrace{\frac{d \ln(\bar{N} - I)}{dI}}_{\text{Reallocation Effect}}.$$

The productivity effect can be expressed in terms of productivities and prices,

$$\frac{d \ln w_\ell}{dI} = \underbrace{\ln\left(\frac{w_\ell}{\gamma_L(I)}\right) - \ln\left(\frac{r}{\gamma_m(I)}\right)}_{\text{Productivity Effect}} - \underbrace{\frac{1}{\bar{N} - I}}_{\text{Reallocation Effect}}. \quad (18)$$

Thanks to this manipulation, we can see that the productivity effect is greater the greater the difference between the cost of producing task I with low-skill labor and with capital. In other words, the greater the cost saved thanks to the automation adoption, the greater the increase in productivity. Thus, whether the low-skill wage decreases in the short run depends on which of the two effects is bigger, which in turn depends on the comparative advantage structure and on the magnitude of the shock. With the chosen calibration for this model, the reallocation effect is stronger than the productivity effect in the short run. In the long run, the accumulation of capital and the increase in the share of agents with a college education affect the transition of wages. The increase in capital tends to increase output and therefore both wages, while the increase in the number of educated workers decreases the wage of high-skill labor and increases the wage of the low-skill workers. To understand the difference between the short and the long run, it is important to notice that while the reallocation effect is instantaneous, the implications of the productivity effect change over time as more and more capital is accumulated. For this reason, the reallocation effect can be greater than the productivity effect in the short run, but the opposite can be true in the long run. Hence, for a fixed reallocation of factors, the greater the productivity effect, the greater the difference between the short- and the long run effects.

To analyze the transition of the college premium and the share of educated

workers, it is useful to report the expression of the wage ratio,

$$\frac{w_h}{w_\ell} \propto \frac{N - \bar{N}}{\bar{N} - \uparrow I} \cdot \frac{(1 - S_h)}{S_h},$$

and the problem of a new-born worker,

$$v_t^n(a) = \max \left\{ \uparrow \mathbb{E}_\varepsilon \left\{ v_t^h(a, \varepsilon^h) \right\} - \theta(a), \quad \downarrow \mathbb{E}_\varepsilon \left\{ v_t^\ell(a, \varepsilon^\ell) \right\} \right\}.$$

Right after the shock, the immediate increase in the college premium implies an increase in the expected lifetime utility of high-skill workers relative to low-skill workers. For this reason, more agents decide to invest in education (increase in S_h) thus implying a decline in the college premium. However, there is another force that tends to increase the share of high-skill agents: as aggregate capital and per capita capital increase, also the average new-born worker becomes richer. As the cost of education is constant in the model, *more* agents can afford to get education. This last point explains why, in the final steady-state, the share of high-skills is higher than its value in the initial steady-state despite the college premium being lower. The increase in the interest rate increases wealth inequality, as the agents who own more capital benefit more from this increase. Also, the sudden jump in the college premium, allows college-educated workers to accumulate more capital. This boost in wealth inequality, combined with the increase in wage inequality implies a permanent increase in the spread of the consumption distribution, measured with the standard deviation.

In Figure 7 I further focus on the effect of automation on inequality. To show how the total income and consumption distribution react to the sudden substitution between low-skill labor and capital in production, I report, for each distribution, the relative change with respect to the initial steady-state value of each percentile. In these graphs, each line represents the change of percentiles in a *given period* after the shock. The blue line depicts the change at $t = 3$, which is almost immediately after the shock. To understand what happens, recall that the wage of the uneducated workers decreases immediately but the interest rate increases. As a consequence, about half of the population in the model economy experiences a decrease in total income when tasks are automated. This, is even though the fraction of agents without a college degree is 82% in the initial steady-state (see Table 3). The reason behind this is that uneducated workers

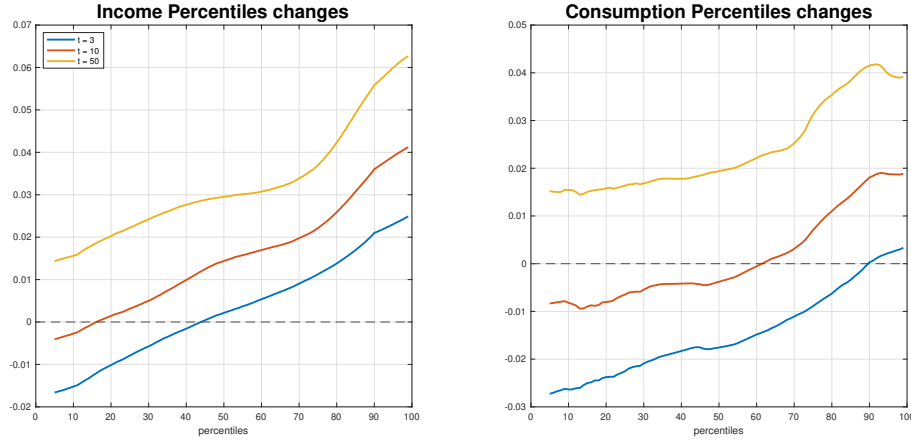


FIGURE 7: Percent variation of percentiles with respect to the initial steady-state value. Each line represents the variation a given number of years after the shock hits the economy. Left panel, total income distribution. Right panel, consumption distribution.

who had a lucky series of shocks and managed to accumulate relatively more wealth, do not see their total income decrease, as the increase in the capital income *compensates* the decline in their return from the labor market. This highlights the importance of departing from the representative household model when studying the effect of technology on total income and consumption. In that model, as automation increases output, even if labor income declines, the household is compensated with the increase in capital income. As a consequence, consumption and welfare necessarily increase. This is not true anymore with heterogeneous agents: only a small fraction of the agents who see their labor income drop benefit from greater capital income. The implication for the consumption distribution can be seen in the right panel of the same figure. Right after the shock (blue line), all households below the 90th percentile experience a decrease in consumption. The decrease in consumption is greater than the one in the income distribution because agents are taking advantage of the high interest rate and, therefore, postpone their consumption. As time goes by, both distributions “shift to the right” and approximately after 50 years every percentile of both distributions is at a higher value. To summarize the effect of automation, we observe that in the short run some agents lose, in the long run-everybody is better off, but polarization has increased.

In Figure 8, I report the transition after an introduction of tasks in which labor has a comparative advantage with respect to capital, i.e., an increase in N .

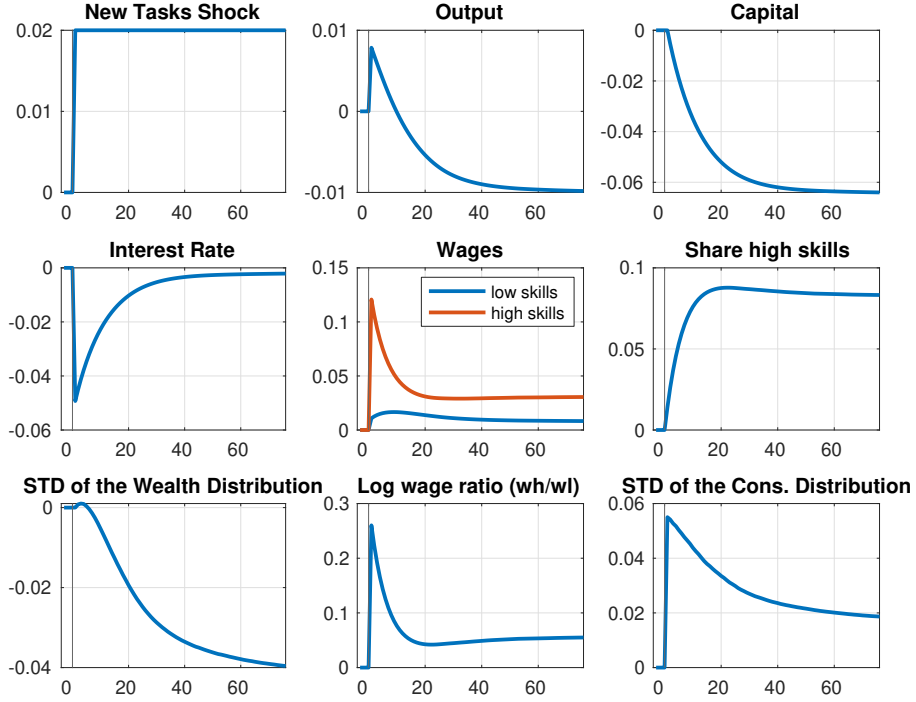


FIGURE 8: Transitional dynamics between the initial, calibrated steady-state and a final steady-state in which the highest-indexed task in the economy, N , increases by 2%. All reported variables are normalized to zero in the initial steady-state.

In the short run, the introduction of new tasks increases productivity, and, consequently output increases. However, the production of the final good becomes more labor-intensive and the relative demand for capital decreases. The effect on the absolute demand for capital depends on the interaction between the boost in productivity and the decrease in the relative demand for capital; with the chosen calibration the demand for capital decreases, as can be observed in the transition of the interest rate. The decrease in the interest rate implies that the aggregate level of capital starts decreasing, in turn implying a decrease in the production of the final good. Both wages increase because of the increase in productivity; however, the wage of educated workers increases more, as they also benefit from the increase in the relative demand for skilled labor. By looking at the following equation,

$$\frac{w_h}{w_\ell} \propto \frac{\uparrow N - \bar{N}}{\bar{N} - I} \cdot \frac{(1 - S_h)}{S_h},$$

it is clear why the college premium increases in the short run. Along with the transition, as more agents choose to invest in education given the increased pre-

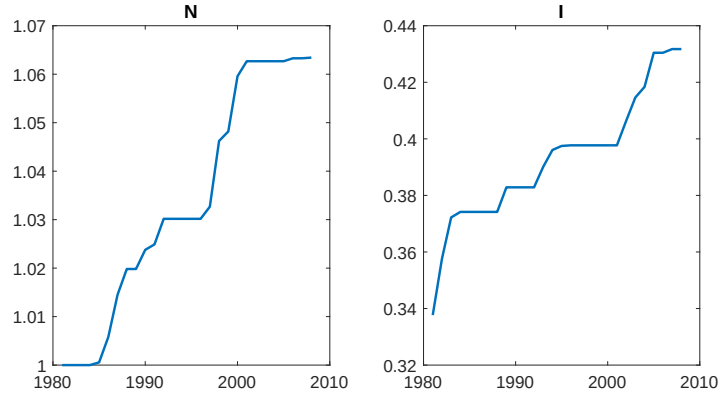


FIGURE 9: Estimated sequences of the technology parameters N —left panel—and I —right panel.

mium, the gap between the wages of high- and low-skill workers declines until its final steady-state value. The effect on wealth inequality depends, as before, on the interaction between the effect on the return of capital and wage inequality. The increase in wage inequality dominates in the short run, implying an increase in wealth inequality which, however, decreases along with the transition because of the lower interest rate and the decreasing college premium. For similar reasons, the standard deviation of the consumption distribution increases when new tasks are introduced and reaches its maximum right after the shock. After that, it starts to decline.

It is crucial to underscore that the transitions discussed in this section are solely intended to elucidate the mechanisms within the model and should not be misconstrued as direct comparisons to empirical data. Notably, the processes of automation and task introduction are intricately linked, unfolding in a continuous and mutually dependent way. To validate the model effectively, it becomes imperative to examine a transition in which both components of technological change—automation and task introduction—progress concurrently over time. This validation process is addressed in the subsequent sections of this paper.

3.3 Estimation of the Technological Change

In the previous section, the shocks are instantaneous and their magnitudes are chosen ad hoc to have the clearest possible dynamics. In this section, instead, I explain how I estimate from the data the sequences of the two technology

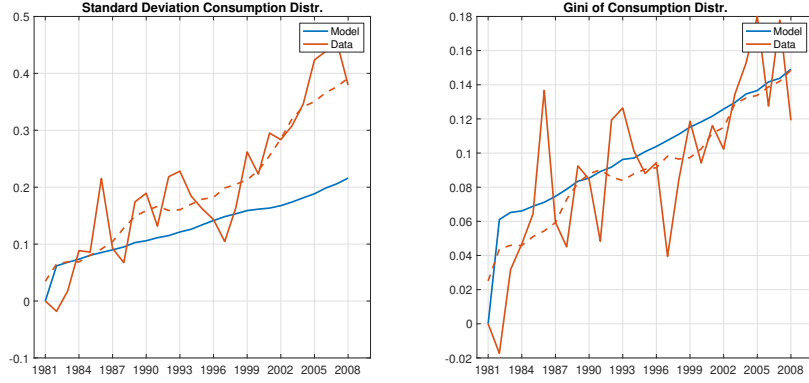


FIGURE 10: Evolution over time of the standard deviation and the Gini coefficient of the consumption distribution. The model-generated series are contrasted with the data counterparts. All values are normalized to zero in 1981. For illustrative purposes, I also report the 10-year moving average for the data series.

variables, $\{I_t, N_t\}$. In the following section, I use these estimated sequences to compute the transition that I then compare with the data. For the estimation of I_t and N_t , I use the series of the labor share as reported by the BEA (see Figure 4) and the expression that links, in the model, these two variables with the labor share, $I_t = N_t - (\text{LABOR SHARE})_t$. Given the initial values for I_0 and N_0 I adopt a similar technique as in [Acemoglu and Restrepo \(2019\)](#) which in turn relies on the theory developed in [Acemoglu and Restrepo \(2018b\)](#): in a model with endogenous technological change, in a given period, is either profitable to develop technologies which are labor-intensive or automation technologies with the purpose of substituting labor in a given set of tasks. With this reasoning in mind, I assume that in a given period there are three possibilities: an increase in I , an increase in N , or no technological change. Following this logic, if the labor share increases I impute this increase to the introduction of new tasks, if it decreases, to the automation of tasks. This computation results in the sequences reported in Figure 9.

3.4 Transitional dynamics of the calibrated economy

In this section, I compute the transition of the model economy implied by the estimated sequences and then compare this transition with real data spanning from 1981 to 2008. To simulate a balanced growth path, I extend the estimated

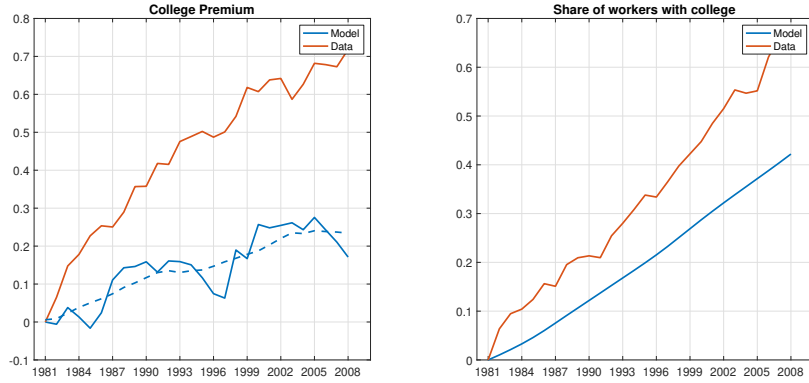


FIGURE 11: Evolution over time of the college premium and the share of workers with a college degree. The model-generated series are contrasted with the data counterparts. All values are normalized to zero in 1981. For illustrative purposes, I also report the 10-year moving average for model-generated college premium.

sequences with linear trends for 50 years. After this period, the technology parameters remain constant. Because of future discounting and the dying probability in every period, what happens in the first 30 years after the initial steady-state—which is the period under study—is not affected by what happens in a so distant future.

In Figure 10, I contrast the model-generated series of consumption inequality with the data. For completeness, I report both the Gini coefficient and the standard deviation of the distribution. The standard deviation implied by the model closely mirrors the distribution spread observed in the data until 1999. At that point, the inequality measured in the data increases relative to the model. The Gini coefficients are pretty close along the period under study: the model is able to generate the 14% increase in the Gini coefficient. Consumption inequality depends on the college premium and on the fraction of workers with a college degree, for this reason, in Figure 11 I also contrast the evolution of these variables with the data. The model is able to explain around 35% of the increase in the college premium and around 63% of the increase in the share of educated workers.

3.5 Effect Decomposition

In this section, I analyze the role played by task automation, the introduction of new tasks, the return of wealth, and education decision in the transition shown

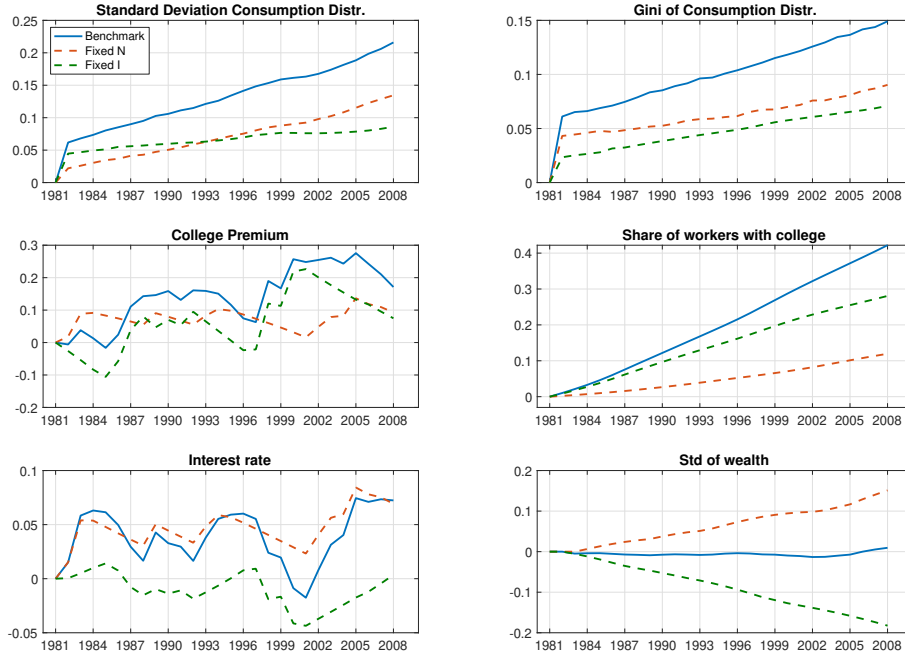


FIGURE 12: This figure compares the benchmark transition with two transitions in which I fix the automation of tasks and the introduction of new tasks.

in the previous section. To understand the quantitative role of these channels, I compute the transition keeping each component fixed to the initial steady-state value.

In Figure 12, I report the transition in which I fix the capital intensity to the initial steady-state value, the transition with no introduction of new tasks, and the benchmark transition for comparison. From this figure, we see how task automation contributes to the increase in inequality. The college premium increases *less* when no tasks are automated and, as a consequence, the share of college workers is also smaller along with the transition. Task automation also contributes to the increase in the return of wealth. The lower return to wealth combined with the lower college premium implies that both measures of consumption inequality are lower in every year under study when no tasks are automated.

In the same figure, we see how also the introduction of new tasks contributes to the increase in consumption inequality. First, the introduction of new tasks increases the college premium. Second, as the relative demand for capital decreases because production becomes more labor-intensive, task introduction decreases the value of the return to wealth. This has the effect of decreasing wealth

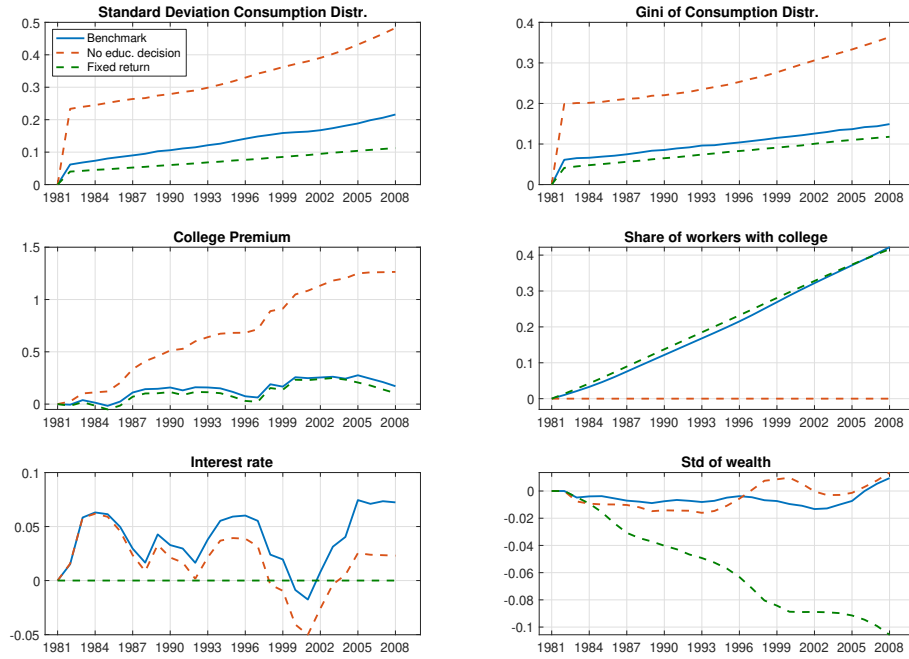


FIGURE 13: This figure compares the benchmark transition with two transitions in which I shut down the educational choice decision and the increase in the return to wealth.

inequality. However, the increase in the college premium dominates and, according to the model, the introduction of new tasks has contributed to the observed increase in consumption inequality between 1981 and 2008.

In Figure 13, I report the transition with a fixed return to wealth and fixed education shares. Similarly to the previous exercises, I fix the interest rate to the initial steady-state value. As the interest rate is an endogenous variable in the model, fixing it implies that along with the transition and in the final steady-state the capital supply does not equal the capital demand. This exercise shows the importance of taking into account heterogeneous capital accumulation when studying the implication of technological change on inequality. Indeed, the increase in the interest rate contributes to the growth of consumption inequality by almost 4%.

In the same figure, I report the transition in which agents do not have the opportunity to choose their education. To this end, I set to zero the probability of dying, d , in the household optimization problem.¹⁵ In this way, the shares of college and non-college-educated agents remain fixed at the initial steady-

¹⁵I also need to change the value of β in order to have comparable time discounting.

state level. When the labor force does not adjust the skill supply, the level of inequality is much higher. The college premium increases dramatically more with respect to the benchmark transition. As the college premium increases, both measures of consumption inequality increase *more* as well. The role of educational choice is therefore to buffer the increase in inequality implied by technological change.

4 Conclusion

I study the relationship between automation and consumption inequality by combining two theoretical frameworks. I use an incomplete market model à la Aiyagari with educational choice combined with a task-based model of production. After calibrating the model to the US economy between 1978 and 1980, I first show that automation decreases the labor income of uneducated workers and that the implied increase in the return to wealth counteracts that drop only for the uneducated rich. As the high-skill workers earn more and have, on average, higher wealth, the increase in the return to wealth widens the gap between the total income of high- and low-skill workers. To summarize, the adoption of automation technology makes some of the agents lose in the short run; every agent is better off in the long run, but the level of inequality is higher.

After estimating the series of automation and new task creation, I compute the model implied transition and contrast this with the data. The model is able to replicate the increase in consumption inequality that took place in the US between 1981 and 2007. Finally, I decompose the effects of various components of the model along with the transition. First, I quantify the role played by the automation of tasks and the introduction of new tasks, highlighting how new tasks for labor can counteract the decrease in labor demand associated with automation but tend to increase polarization in the economy. Finally, I find that both educational choice and the return to wealth channel are quantitatively important in accounting for the increase in consumption inequality. These two channels are not usually included in the analysis of the dynamic effects of automation, and I argue in this paper that they should be taken into account.

A natural use of the calibrated model developed in this paper is to use it for policy analysis. Concerned about the possible effect of massive adoption

of automation technology, a tax on robots has been proposed and evaluated by economists¹⁶ as a way to control and redistribute the benefits associated with task automation. The framework developed in this article provides an ideal laboratory in which to study the dynamic effects of such policies, and this is left for future research.

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¹⁶For example, [Guerreiro et al. \(2017\)](#).

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